

You can copy most of the following directly into **Project_Documentation.docx** and then add screenshots from your Power BI dashboard.

Retail Sales Data Processing and Business Insights Generation

1. Introduction

ABC Retail Solutions collects retail transaction data from multiple sources. Due to inconsistencies such as duplicate records, missing values, invalid quantities, inconsistent product naming conventions, and varying date formats, generating accurate business insights becomes challenging.

The objective of this project is to build an end-to-end data pipeline that performs data ingestion, cleaning, transformation, validation, PII masking, KPI generation, and visualization through Power BI dashboards.

2. Project Objectives

The primary objectives of this project are:

- Ingest retail transaction data from multiple source systems.
 - Identify and resolve data quality issues.
 - Standardize product and category information.
 - Protect customer sensitive information through PII masking.
 - Generate business KPIs and analytical insights.
 - Develop an interactive Power BI dashboard.
 - Provide a scalable and maintainable data processing workflow.
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3. Dataset Description

The project uses three datasets.

Product Details Table

This table acts as the reference dimension table and contains:

- Product ID
- Product Name
- Category
- Standard Product Price

Purpose:

- Product validation
 - Category standardization
 - Missing price replacement
-

Retail Data 1

Contains retail transaction records collected from source system 1.

Important fields include:

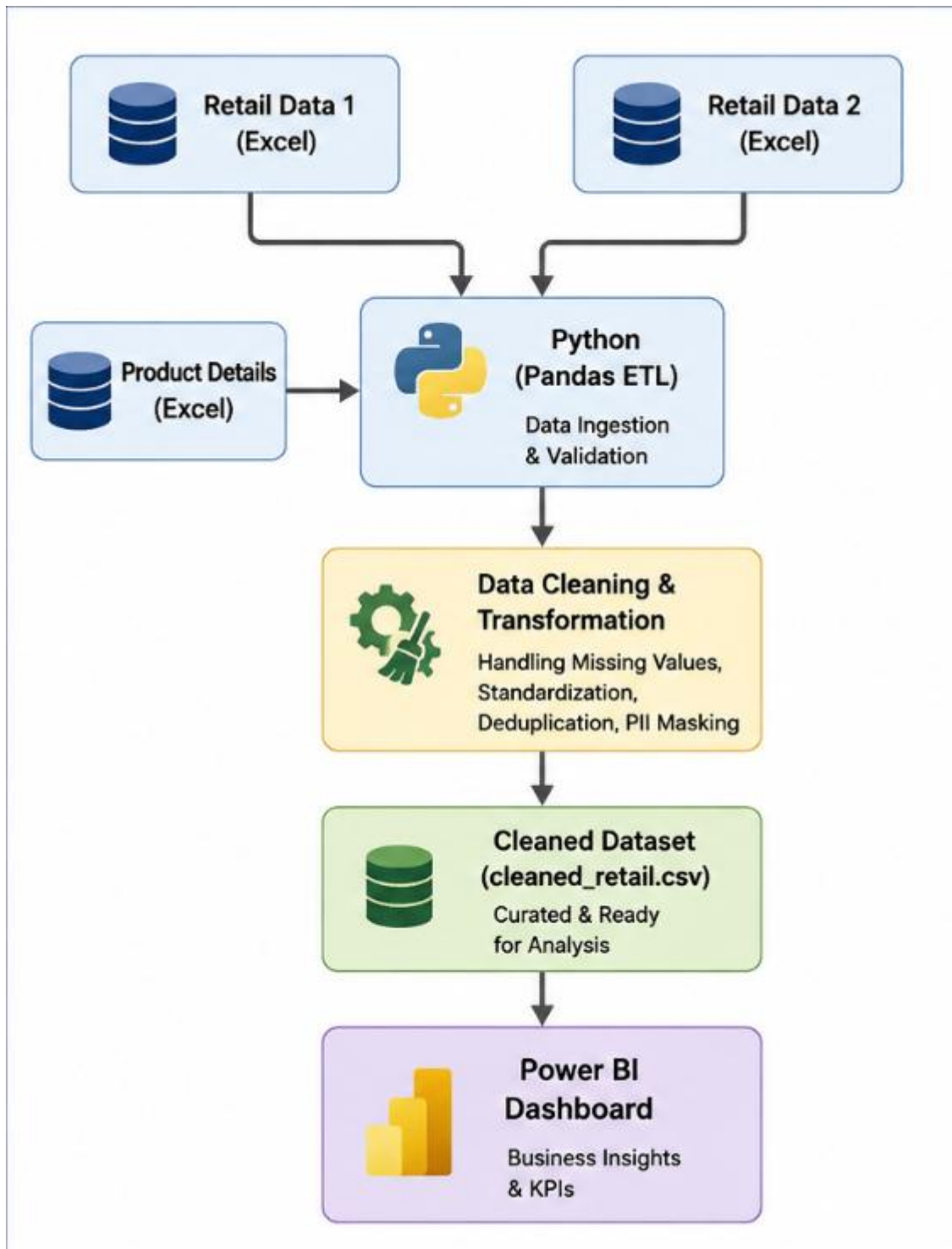
- Transaction ID
 - Customer Information
 - Product Information
 - Quantity
 - Price
 - Discount
 - Payment Method
 - City
 - Transaction Date
-

Retail Data 2

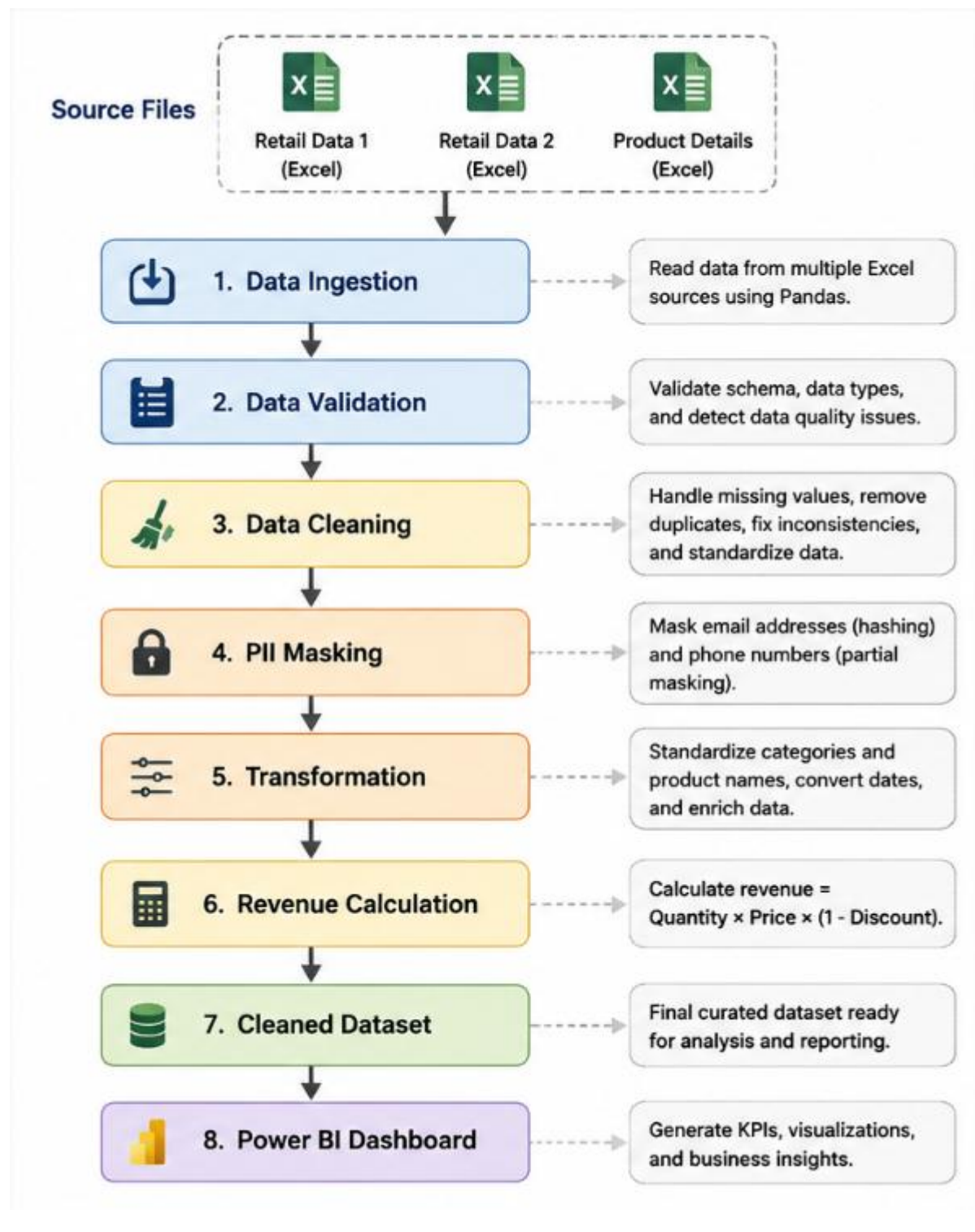
Contains retail transaction records collected from source system 2 and follows the same structure as Retail Data 1.

4. Solution Architecture

Architecture Diagram



5. Data Flow Diagram



6. Data Cleaning Strategy

Several data quality issues were identified and resolved.

6.1 Duplicate Removal

Duplicate transaction records were identified and removed to ensure accurate reporting and KPI calculations.

Action Taken:

- Duplicate rows were detected using transaction-level information.
 - Duplicate records were removed.
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6.2 Missing Value Handling

Missing values were identified in multiple columns, particularly in price fields.

Action Taken:

- Missing prices were replaced using the standard price available in the Product Details table.
 - Invalid or incomplete records were reviewed and corrected wherever possible.
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6.3 Product Name Standardization

Different naming conventions existed for the same product.

Examples:

Laptop

laptop

LAPTOP

Action Taken:

- Converted product names into a consistent format.
 - Removed leading and trailing spaces.
-

6.4 Category Standardization

Category values were inconsistent.

Examples:

Electronics

electronics

ELEC

Furniture

furniture

FURN

Action Taken:

- Created mapping rules.
 - Converted all category values into a standard format.
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6.5 Date Standardization

Multiple date formats were present.

Action Taken:

- Converted all dates into a standard datetime format.
 - Invalid dates were handled appropriately.
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6.6 Quantity Validation

Invalid quantities were identified.

Examples:

Quantity = 0

Quantity = -5

Action Taken:

- Removed records with quantity less than or equal to zero.
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7. PII Masking Strategy

Customer-sensitive information was protected to ensure privacy and compliance.

Email Protection

Action Taken:

- Customer email addresses were hashed using SHA-256 hashing.

Example:

Original:

john@example.com

Masked:

3d5f2a...

Phone Number Protection

Action Taken:

- Phone numbers were partially masked.

Example:

Original:

9876543210

Masked:

98*****10

8. Transformation Logic

The following transformations were performed.

Product Enrichment

The retail transaction dataset was joined with the Product Details table using Product ID.

Purpose:

- Category validation
 - Product standardization
 - Price correction
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Revenue Calculation

Revenue was calculated using:

Revenue = Quantity × Price × (1 - Discount)

The calculated revenue field serves as the primary metric for reporting and analysis.

9. Data Model

A star schema approach was followed.

Product Details (Dimension)

|

v

Retail Transactions (Fact Table)

Dimension Table:

- Product Details

Fact Table:

- Retail Transactions
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10. KPIs Generated

The following KPIs were developed.

Total Revenue

Measures total sales revenue generated.

Revenue by Category

Measures revenue contribution from each product category.

Revenue by City

Measures revenue generated across different cities.

Total Orders

Measures the number of unique transactions.

Total Customers

Measures the number of unique customers.

Average Order Value

Formula:

Average Order Value = Total Revenue / Total Orders

Top Selling Products

Identifies products generating the highest revenue.

Payment Method Analysis

Measures revenue distribution across payment methods.

11. Power BI Dashboard Design

The dashboard was divided into multiple analytical sections.

Executive Summary

Contains:

- Total Revenue
- Total Orders
- Total Customers
- Average Order Value

Visuals:

- KPI Cards
 - Revenue Trend Chart
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Product Analysis

Contains:

- Top Selling Products
- Revenue by Product

Visuals:

- Bar Charts
 - Product Ranking Charts
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Category Analysis

Contains:

- Revenue by Category
- Category Contribution

Visuals:

- Donut Charts
 - Column Charts
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Regional Analysis

Contains:

- Revenue by City
- Regional Performance

Visuals:

- Bar Charts
 - Geographic Visualizations
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12. Business Insights

Some key observations obtained from the analysis include:

- Electronics generated the highest revenue contribution.
 - Certain cities contributed significantly higher sales than others.
 - A small group of products generated a major share of total revenue.
 - Payment preferences varied across customers.
 - Seasonal patterns were observed in monthly revenue trends.
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13. Assumptions

The following assumptions were made during implementation:

1. Negative quantities are invalid and removed.
 2. Missing prices are replaced using the standard product price.
 3. Product ID uniquely identifies a product.
 4. Duplicate transaction records represent data quality issues.
 5. Masked customer information is not required for reporting purposes.
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14. Conclusion

This project successfully implemented an end-to-end retail data engineering pipeline involving data ingestion, cleansing, transformation, validation, PII protection, KPI generation, and dashboard visualization. The final solution provides reliable business insights and supports data-driven decision-making through an interactive Power BI reporting platform.

1.24bn

Total Revenue

8K

Total Orders

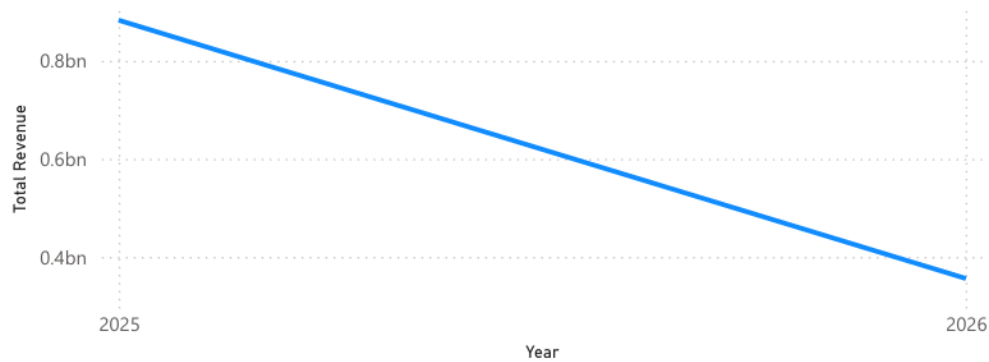
2K

Total Customers

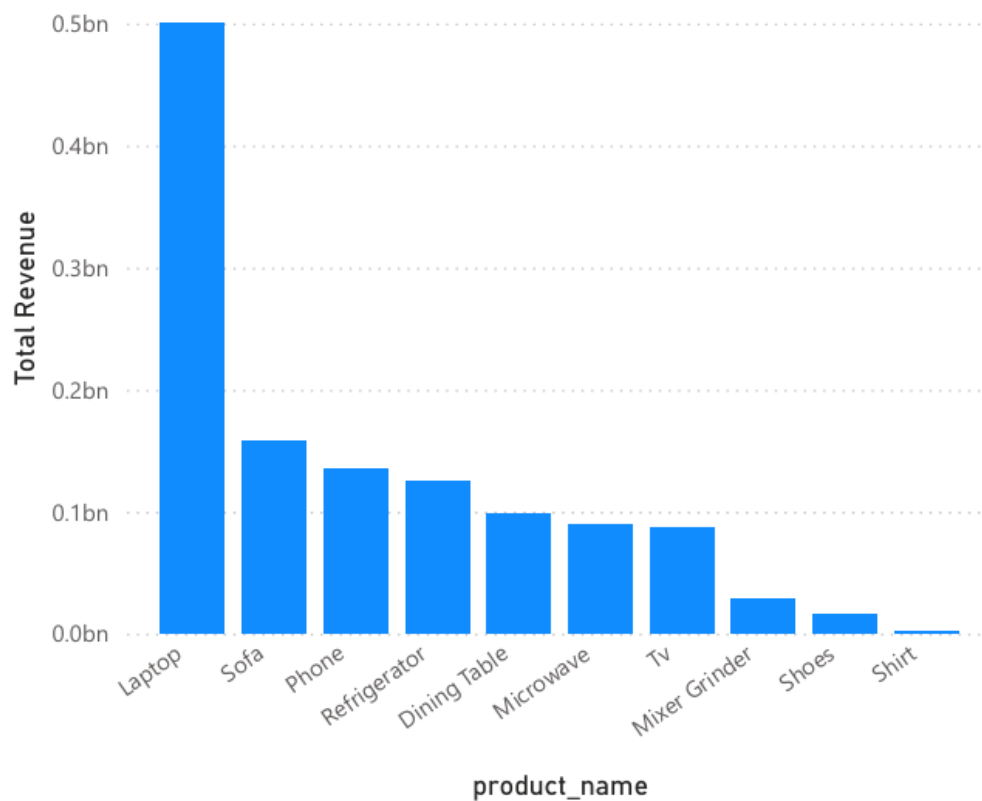
156.57K

Average Order...

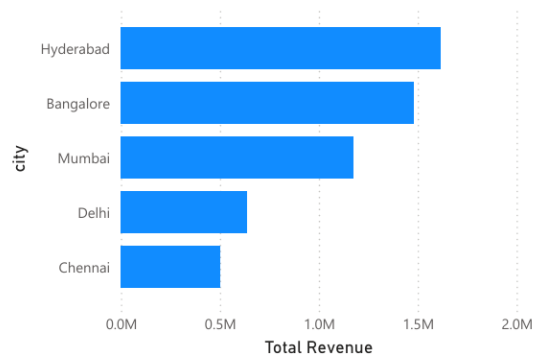
Total Revenue by Year



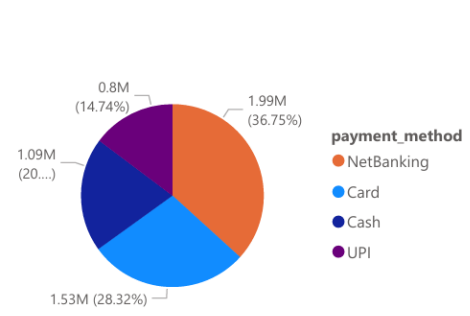
Top 10 Products by Revenue



Total Revenue by city



Total Revenue by payment_method



city

- ☐ Bangalore
- ☐ Chennai
- ☐ Delhi
- ☐ Hyderabad
- ☐ Mumbai

category

- ☐ Clothing
- ☐ Electronics
- ☒ Furniture
- ☐ Home Appliances

transaction_date